OXFORD
INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL GCSE

Biology

9201/1 Paper 1
Mark scheme

9201
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Version/Stage: 1.0 Final

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way.

As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

2.1 In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.

2.2 A bold **and** is used to indicate that both parts of the answer are required to award the mark.

2.3 Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

Any wording that is underlined is essential for the marking point to be awarded

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of error / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	**Response**	**Marks awarded**
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	**Response**	**Marks awarded**
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of ‘it’

Answers using the word ‘it’ should be given credit only if it is clear that the ‘it’ refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(…..) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do <u>not</u> accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

4. Level of response marking instructions

Extended response questions are marked on level of response mark schemes.

- Level of response mark schemes are broken down into levels, each of which has a descriptor.
- The descriptor for the level shows the average performance for the level.
- There are two marks in each level.

Before you apply the mark scheme to a student's answer, read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

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Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO/Spec. Ref.
01.1	<table><tr><th></th><th>Bacterial cell</th><th>Animal cell</th></tr><tr><td>Cell wall</td><td>✓</td><td></td></tr><tr><td>Nucleus</td><td></td><td>✓</td></tr><tr><td>Cell membrane</td><td>✓</td><td>✓</td></tr><tr><td>Cytoplasm</td><td>✓</td><td>✓</td></tr></table>		1 1 1 1	AO1 3.1.1a/c
01.2	mitochondria		1	AO1 3.1.1a
01.3	any **one** from: • chloroplast • cell wall • [permanent] vacuole	ignore chlorophyll	1	AO1 3.1.1b
01.4 View with 01.3	chloroplast – to make food **or** for photosynthesis **or** to absorb light **OR** cell wall – strengthens cell (1) **OR** [permanent] vacuole – holds cell sap (1)	 allow supports the cell **or** keeps cell turgid	1	AO1 3.1.1b
01.5	$\frac{20}{1000}$ 0.02 (mm)	an answer of 0.02 (mm) scores 2 marks	1 1	AO2 3.1.1b
Total			**9**	

Question	Answers	Extra information	Mark	AO/Spec. Ref.
02.1		must link adaptation with correct description		AO2 3.3.2d
	spikes		1	
	to deter predators	allow not eaten by predators	1	
	OR			
	mottled/striped (1)	allow description		
	so less likely to be seen by predators/prey (1)	allow not eaten by predators allow camouflage from predators/prey		

02.2	large ears hear predators/prey **or** large surface area for heat loss **OR** eyes at front of head **or** large eyes (1) see predators/prey [at night] (1) **OR** Fur (1) keep warm/insulated **at night** (1)	must link adaptation with correct description	1 1	AO2 3.3.2d

02.3	half		1	AO2 3.4.3a

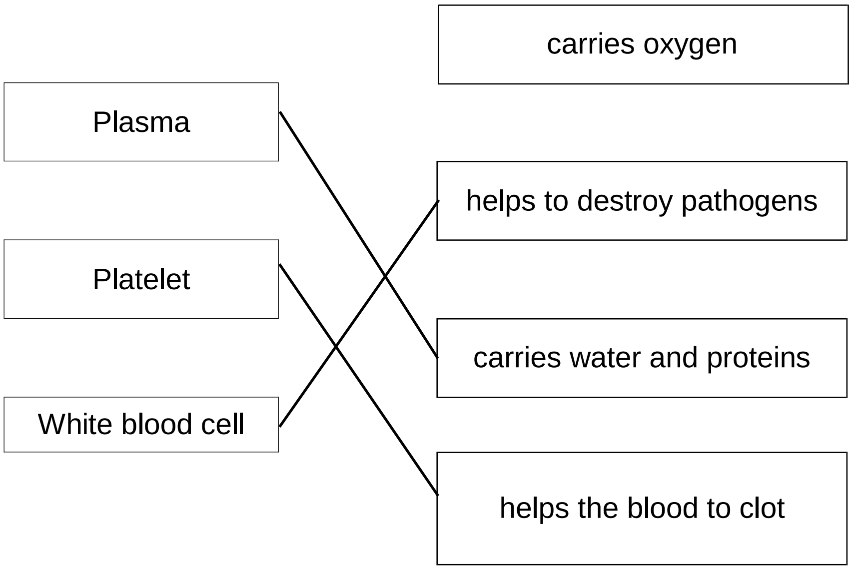
02.4	brain		1	AO1 3.4.3e

02.5	ADH	allow antidiuretic hormone	1	AO1 3.4.3e

02.6	(aerobic) respiration	do not allow anaerobic respiration	1	AO1 3.2.6c

02.7	deamination (of amino acids) ammonia formed (ammonia) converted into urea	allow amino group removed	1 1 1	AO1 3.4.3e

Total			**11**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
03.1	 If more than one line is drawn from a component then no mark.		3	AO1 3.2.3l/m/o/p
03.2	**Pulmonary** artery		1	AO1 3.2.3i
03.3	Stronger / more (muscle) contraction (to create) higher (blood) pressure (to) pump blood all around the body	comparative must be given at least once	1 1 1	AO2 3.2.3b/h
03.4	135		1	AO2 3.2.3s

03.5	**Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5-6	AO3 3.2.2g
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3-4	AO3 3.2.2s
	Level 1: Relevant points are made. They are not logically linked.	1-2	AO2 3.2.2s
	No relevant content	0	
	Indicative content allow converse arguments **Advantages of Jarvik Heart** • no need to wait for the transplant • surgery less drastic **Disadvantages of Jarvik Heart** • more chance of blood clots and strokes • only lasts up to 5 years • the mechanical motor can fail • only treats one heart condition **Comparisons** • mechanical only lasts 5 years **but** a heart transplant last much longer, 20 years • mechanical heart only treats one heart problem **but** heart transplants treat many • mechanical heart readily available **but** a heart for transplant may not be **Additional detailed knowledge** • no chance of being rejected by the body [for mechanical heart] • do not need to take anti-rejection drugs [for mechanical heart] • may cause an immune response, ref to foreign antigens [for heart transplant]		

Total			**14**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
04.1	**A**: salivary glands **B**: liver		1 1	AO1 3.2.4a
04.2	(in foldings of membranes or (micro)villi provide) large(r) surface area for faster diffusion **OR** (microvilli) are thin (1) to give a short diffusion pathway (1)	 ignore easier (diffusion) allow faster absorption	1 1	AO2 3.1.5i/b
04.3	fatty acid **and** glycerol	both required for the mark in either order	1	AO1 3.2.4c
04.4	emulsifies fats to give a large surface area for enzyme action/digestion **OR** neutralises (stomach) acid (1) to give optimum / alkaline conditions for lipase/enzymes to work (effectively) (1)	allow description of emulsification ignore breakdown of fats (unqualified)	1 1	AO1 3.2.4c

04.5	any **two** from: • add same volume of lipid • add same volume of lipase • same concentration of lipid • same concentration of lipase • leave solutions (for set time) in water bath before mixing • repeats at each temperature	Allow 1 mark for amount of lipid **or** lipase if neither of these is given Allow 1 mark for concentration of solution if neither of these is given	2	AO4 3.2.4b
04.6	(468 + 482) ÷ 2 475	an answer of 475 scores 2 marks allow 1 mark for an answer of 397	1 1	AO2 3.2.4b
04.7	any **two** from: • mean results at 40°C and 50°C are similar • have not tested temperatures close to 45°C • anomalous result at 30°C	allow they have not tested between 30°C and 40°C **or** 40°C and 50°C	2	AO4 3.2.4b
04.8	(Enzyme) denatured **or** shape of active site changed	allow description of shape change **do not** allow killed	1	AO2 3.2.4b
Total			**14**	

Question	Answers	Extra information	Mark	AO/Spec. Ref.
05.1	any **three** from: • uncontrolled/abnormal growth of cells • which can invade other/neighbouring tissues • (cells) spread to other organs or in the blood or to the rest of the body • form secondary tumours in other organs	ignore ref to cancer allow can metastasize	3	AO1 3.5.2n
05.2	destroys/no receptors between 8 and 25 arbitrary units.		1	AO3 3.4.2c 3.4.1a
05.3	any **two** from: • where there are no receptors light will not be detected • so no impulses are sent to the brain • so no image is formed there **or** so there is a gap in the vision/visual field		2	AO3 3.4.2c 3.4.1a/d
05.4	only expressed if the dominant/other (allele) is not present **or** only expressed if 2 (recessive alleles) are present	allow will not be expressed if the dominant (allele) is present	1	AO1 3.5.3g

05.5	parental genotypes correctly identified (on genetic diagram) as: Rr **and** Rr (correct parental genotypes) **Rr** <u>and</u> **Rr** **or** (correct gametes) **R** and **r** <u>and</u> **R** and **r**		1	AO1 3.5.3g
	offspring genotypes correctly derived (may be shown in a diagram, *allow ecf from parental gametes)* **RR Rr Rr rr**		1	
	identification of **rr** as having retinoblastoma eg rr – has retinoblastoma		1	
	probability correct as 0.25	allow 25 / 25% / $\frac{1}{4}$ / 1 in 4 / 1 out of 4 or 1:3 ratio do **not** allow 3:1 or 4:1	1	

Total			**11**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
06.1	pancreas		1	AO1 3.4.5c
06.2	any **two** from • lowers blood glucose levels • by allowing glucose to move (from blood) into (liver) cells • to store as/convert into glycogen.	must be clear that glucose moves **into** the cells and not out allow glucose moving into liver/muscles	2	AO1 3.4.5a/b
06.3	$\frac{0.8}{60}$ $0.01\dot{3}$ $mg/dm^3/min$ **OR** 0.8 (1) In 1 hour = 0.8 (1) $mg/dm^3/hr$ (1)	an answer of $0.01\dot{3}$ scores 2 marks allow $mg\ dm^{-3}min^{-1}$	1 1 1	AO2 3.4.5b/c

06.4	low/little glucose in blood after 2.5 – 3 hours	allow any correct ref to data	1	AO2 3.4.5c 3.2.6b/c AO3
	(low/little glucose) for respiration		1	
	to release energy	do **not** allow to produce energy	1	
	blood glucose levels cannot be brought back to normal	allow blood glucose levels stay low	1	
	because glycogen not converted glucose		1	

Total			**11**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
07.1	detect the chemicals	allow chemical stimuli	1	AO2 3.4.2b
07.2	any **two** from: • increased surface area (for contact with receptors) • can sense/detect molecules/chemicals from different areas/places • to identify position/direction of prey/mate/shelter etc		2	AO2 3.4.2c
07.3	any **two** from: • to brain and not to spinal cord • no synapses • no relay neurone (involved) • no motor neurone (involved)	allow converse if made clear	2 max	AO2 3.4.1d
07.4	avoid danger/damage (to the body)		1	AO2 3.4.1c
07.5	chemical released from one neurone which passes/diffuses to the next neurone	allow neurotransmitter	1 1	AO1 3.4.1d

07.6	impulse won't reach motor neurone so intercostal muscles/diaphragm muscle won't contract (so) the rib cage/diaphragm will not move (and therefore) air will not be moved into the lungs	 allow no increase in volume **or** decrease in pressure	1 1 1 1	AO2/AO1 3.4.1d 3.2.5b

Total			**12**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
08.1	deforestation reduces numbers the most or poaching the least	allow use of figures	1	AO3 3.3.4a/c
	protected areas only increasing numbers		1	
	overall human activities reduce wildebeest numbers		1	
08.2	(populations became) geographically isolated	allow described	1	AO1 3.6.2c
	genetic variation within populations	allow wide range of alleles or mutations	1	
	alleles/genes/characteristics which help the organism to survive are selected	**do not** allow 'adapt to survive' allow same idea described in natural selection	1	
	these alleles/genes are passed onto offspring	allow mutations passed on	1	
	populations so different that breeding to produce fertile offspring is no longer possible	allow cannot successfully interbreed	1	
Total			**8**	